

Theory Lectures on Manufacturing Process (NMES 101)

Riser design and casting defects

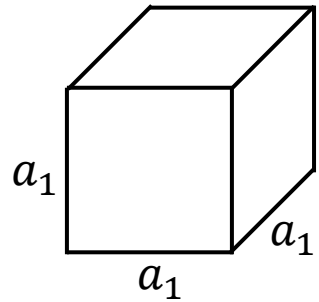


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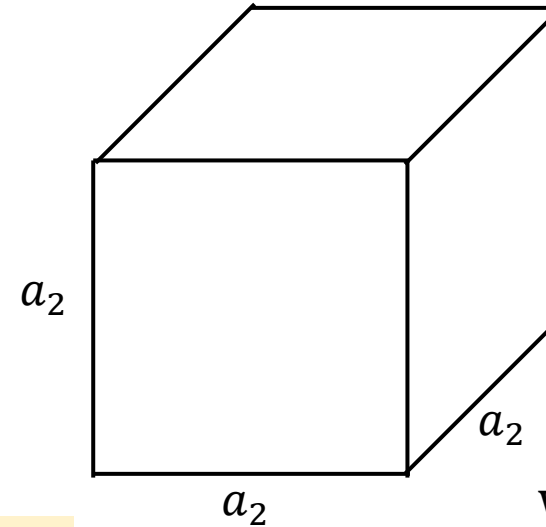
Content prepared for theory lecture for
1st Year BTech level students for common
course on Manufacturing Processes.
Each lecture duration $\approx$ 45 min



Problem: A cube shaped casting solidifies in 5 minutes. Determine the solidification time of a cube of same material which is 8 times heavier.



Volume, $V_1 = a_1^3$
Surface Area, $A_1 = 6a_1^2$



Volume, $V_2 = a_2^3$
Surface Area, $A_2 = 6a_2^2$

$$V_2 = 8V_1 \Rightarrow a_1^3 = 8a_2^3 \Rightarrow a_2 = 2a_1$$

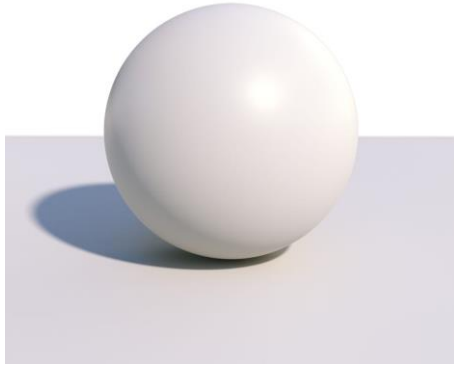
Volume, $V_2 = a_2^3 = 8a_1^3$
Surface Area, $A_2 = 6a_2^2 = 24a_1^2$

Apply Chvorinov's Rule of solidification time

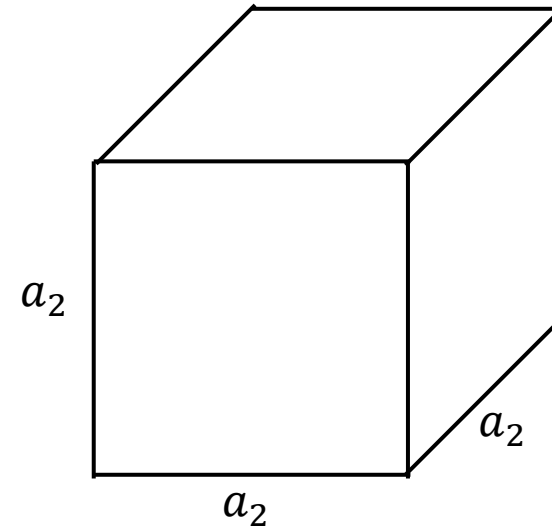
$$\frac{t_2}{t_1} = \frac{\left(\frac{V_2}{A_2}\right)^2}{\left(\frac{V_1}{A_1}\right)^2} = \left(\frac{V_2 A_1}{A_2 V_1}\right)^2 = \left(\frac{8a_1^3 \times 6a_1^2}{24a_1^2 \times a_1^3}\right)^2 = 4$$

$$t_2 = 4t_1 = 20 \text{ min}$$

Problem: Two cast products have the same surface area. One is sphere and the other one is the cube.
Determine the ratio of the solidification time of the sphere and the cube.



$$V_1 = \frac{4}{3}\pi r_1^3$$
$$A_1 = 4\pi r_1^2$$



$$V_2 = a_2^3$$
$$A_2 = 6a_2^2$$

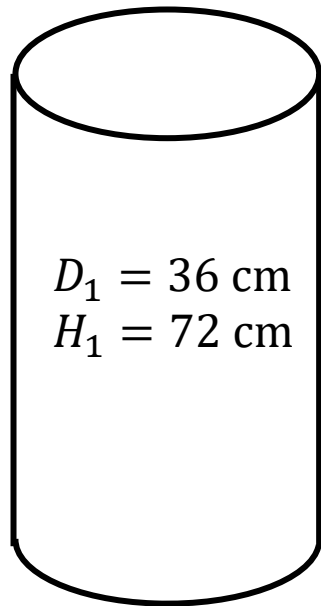
$$A_1 = A_2$$

$$\Rightarrow 4\pi r_1^2 = 6a_2^2$$

$$\Rightarrow \left(\frac{r_1}{a_2}\right) = \sqrt{\frac{6}{4\pi}}$$

$$\left(\frac{t_1}{t_2}\right) = \left(\frac{V_1 A_2}{A_1 V_2}\right)^2 = \left(\frac{V_1}{V_2}\right)^2 = \left(\frac{4\pi r_1^3}{3a_2^3}\right)^2 = \left(\frac{4\pi}{3}\right)^2 \left(\sqrt{\frac{6}{4\pi}}\right)^6 = \frac{6}{\pi}$$

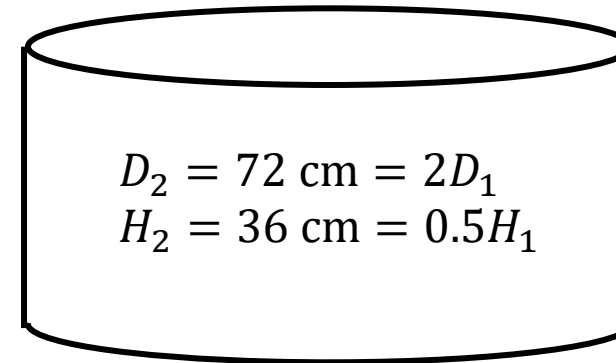
Problem: Calculate the ratio of solidification time of the two cylindrical risers made of steel where one of them is having 36 cm diameter and 72 cm height whereas the other one has 72 cm diameter and 36 cm height .



$$V_1 = \frac{\pi D_1^2 H_1}{4}$$

$$A_1 = \pi D_1 H_1 + \frac{\pi D_1^2}{4}$$

Note: Circular bottom surface of the riser interfaces with the cast metal. So heat loss for solidification cannot occur through this bottom surface.



$$V_2 = \frac{\pi D_2^2 H_2}{4}$$

$$\therefore V_2 = \frac{\pi 4D_1^2 0.5H_1}{4} = 2V_1$$

$$A_2 = \pi D_2 H_2 + \frac{\pi D_2^2}{4}$$

$$\left(\frac{t_1}{t_2}\right) = \left(\frac{V_1 A_2}{A_1 V_2}\right)^2 = \left(\frac{A_2}{2A_1}\right)^2 = \left(\frac{12214.5}{2 \times 9160.88}\right)^2 = 0.44$$



Problem-1: A spherical shaped casting solidifies in 20 minutes. Determine the solidification time of a spherical shaped cast of same material which is 8 times heavier.

Problem-2: A spherical shaped cast is found to solidify within 20 min during sand casting. Calculate the solidification time of a cubic shaped cast product made of same material and has same volume to that of spherical one.

Problem-3: Three different shaped **castings** (i) spherical, (ii) cubic, and (iii) cylindrical with height equals to diameter have same volume. Determine the ratio of their solidification time.

Problem-4: Three different shaped **risers** (i) spherical, (ii) cubic, and (iii) cylindrical with height equals to diameter have same volume. Determine the ratio of their solidification time.

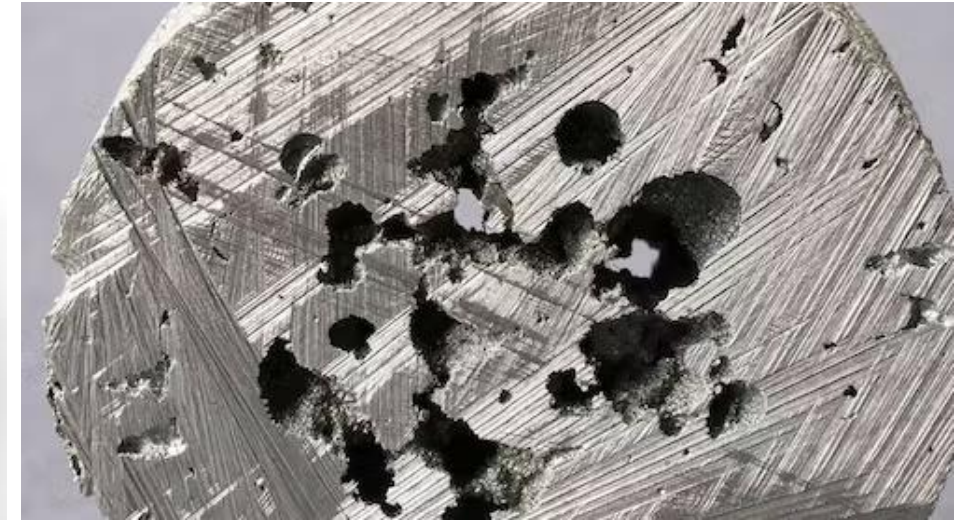


Problem: A casting of a complex shape solidifies in 12 minutes. If all linear dimensions of the casting are doubled, estimate the new solidification time.

Problem: A conical casting solidifies in 30 minutes. A second conical casting is made with the same base radius but double the height. Estimate the solidification time of the second casting.

Problem: A spherical casting of diameter 10 cm solidifies in 20 minutes. What should be the diameter of another spherical casting (same material and mold) so that it solidifies in 80 minutes?.

Problem: A spherical casting and a conical casting (height = $2 \times$ base radius) are made of the same material and have equal volume. Determine the ratio of their solidification times.



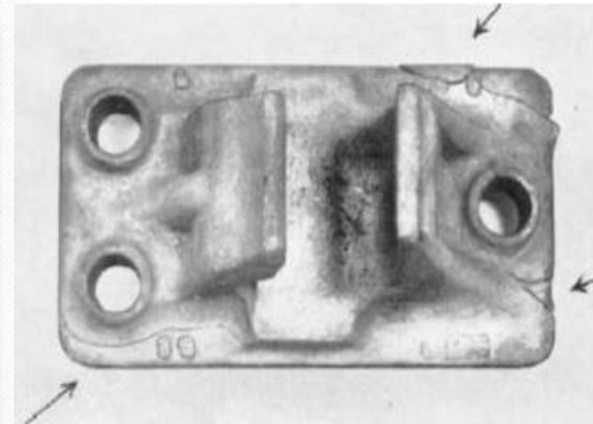
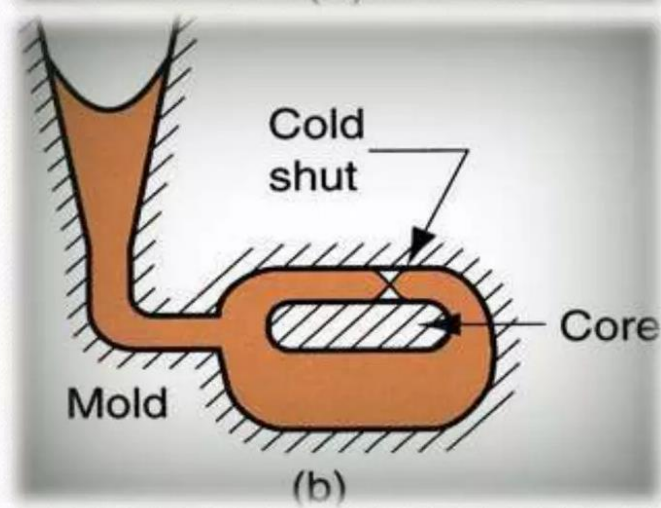
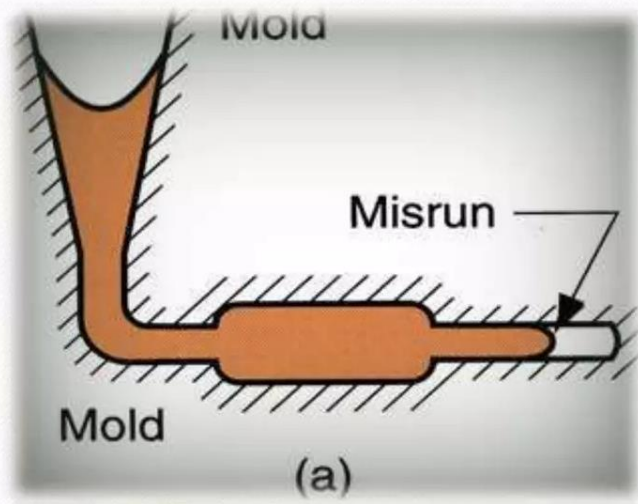
Gas porosity (blowhole / blister) is the formation of rounded or irregular shaped air pockets or voids on the surface or within the casting due to:

- Air entrapment within a casting due to poor design of vents and runner systems.
- Accumulation and entrapment of the dissolved gas (most liquid materials can hold a large amount of dissolved gas).
- Turbulence during pouring/flow of molten metal.
- Vaporization and subsequent entrapment of the water present within the mould sand.
- Improper flow rate and pouring temperature of molten metal.
- Reaction of hot metal with humidity or residual moisture.

Prevention of Gas porosity

- Melt the material in a vacuum or in an environment that has low-solubility for the concerned metal.
- Streamlining the flow of molten metal during pouring as well as flow inside the cavity.
- Reacting the gas with another element to form a compound that will form a dross that floats to the top.
- Drying the mould properly before pouring.
- Adopt proper design of the mould cavity and other accessories.
- Avoid over ramming of sand. Also avoid using fine sand grains (to ensure better permeability of the sand).

Misrun and Cold shut - Casting defects



Cold shut



Misrun

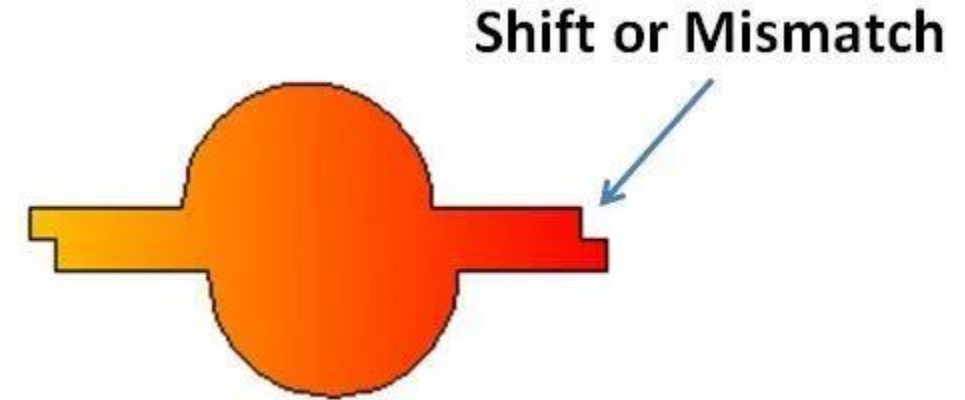
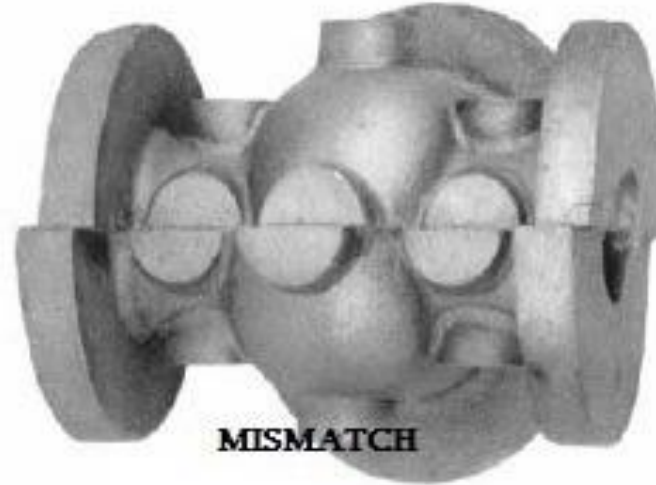
Misrun occurs when the molten metal cannot fill the mold cavity completely and thus leaves improperly filled cavities.

Cold shut occurs when two fronts of liquid metal do not fuse properly in the mold cavity, leaving a weak spot. They both occur mainly due to:

- Lack of fluidity in the molten metal.
- Low pouring temperature.
- Cavity cross-sections are too narrow for the efficient flow of high-viscous molten metal.
- High back pressure from improperly vented mold cavities.
- Very rough mold surface that hinders the flow of molten metal.

Prevention of Misrun and Cold shut

- Adequate pouring temperature (excessive temperatures can again be detrimental, leading to a reaction between the material and the mold – mold may collapse).
- Multiple pouring points to avoid long-length travel of the molten metal poured from one point.
- Appropriate material design – introduce elements to enhance fluidity of molten metal (reduce surface tension through material composition alteration).
- Avoid thin cross-sections in the cavity.
- Smoothen the mould surfaces.

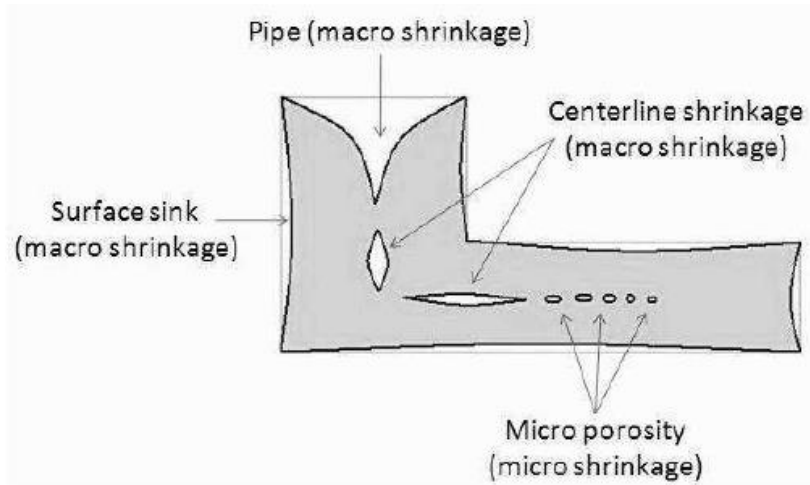


Mismatch occurs due to the misalignment of the lower and upper parts of the mold. This is a serious defect as it mostly leads to the rejection of the product. Primary reasons include:

- The cope and drag (top and bottom halves of the flask) are not properly lined up during mold preparation.
- The clamping pins (pins that hold top and bottom parts of the flask) are worn-out or bent.
- Misalignments of the two halves of the pattern during mold preparation.
- Faulty pattern designs of top and bottom parts.
- Inadequate amount of weight kept on top flask (usually weight is kept for large casting in industrial applications).

Prevention of Mismatch

- Ensure proper match plate pattern alignment and mounting.
- Use the right molding box and closing pins.
- Regular quality control of the flask and pin.
- Appropriate pattern design.
- Keep heavy weight on top for large scale casting.



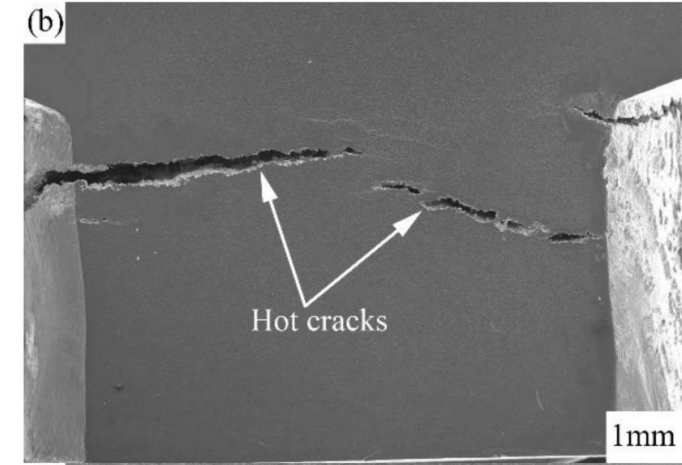
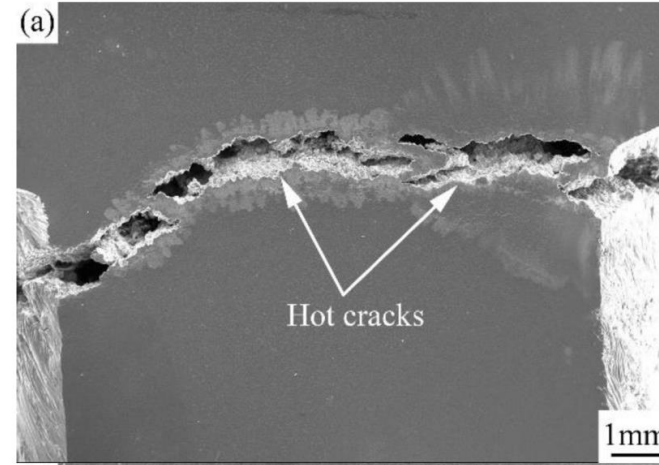
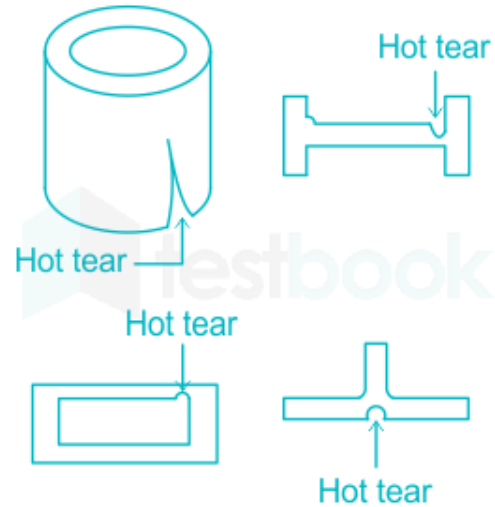
Shrinkage defect occurs when the metal shrinks unevenly, causing it to either distort the shape of the final product or create interior holes. Primary reasons include:

- High temperature gradient across the cast cross-section (uneven solidification).
- Metal is too hot during pouring.
- Inappropriate design of the cavity (pattern) - Sudden change in thickness of sections.
- Insufficient molten metal.
- Riser volume is small or less number of risers.
- High vortex formation or flow instability during molten metal pouring/flow.

Prevention of Shrinkage defect

- Avoid thin sections (make them thicker and then do machining on final product to remove excess metal).
- Use multiple risers of sufficient volume.
- Optimize for continuous flow (runner and gate system) of molten metal at the right temperature.
- Use chills inside the molds to disperse heat faster.

Hot tear / crack - Casting defects



Hot tears appear to be sharp and broken irregular lines within the castings that form during solidification shrinkage. Primary reasons include:

- High thermal-residual stress within the solidifying metal.
- Excess amount of Sulphur/Phosphorus present in cast metal.
- Mold disturbance before complete solidification.

Prevention of Hot tear / crack

- Avoid sharp change in cavity dimension.
- Use chills for controlling solidification rate.
- Use inoculants mixture of Mn/Si/Mg to reduce sulphur/phosphorus content.
- Adopt very slow cooling of the mold metal.

End of casting